

React Basics Notes

1. What is React?

React is a JavaScript library for building user interfaces.

Maintained by Meta (Facebook).

Allows building reusable UI components.

Uses a virtual DOM to efficiently update the UI.

2. Key Features of React

Component-based architecture

One-way data binding

Declarative UI

Hooks for state and lifecycle

JSX syntax (JavaScript + HTML)

3. JSX (JavaScript XML)

JSX lets you write HTML in JavaScript.

```
const element = <h1>Hello, React!</h1>;
```

You can embed expressions using {}.

Components must return a single parent element.

4. Creating a Component

Functional Component:

```
function Welcome(props) {  
  return <h1>Hello, {props.name}</h1>;  
}
```

Arrow Function:

```
const Welcome = ({ name }) => <h1>Hello, {name}</h1>;
```

5. Props in React

Props are used to pass data to components.

Props are read-only.

```
<Welcome name="Alice" />
```

Inside the component: `props.name`

6. State in React

State is used to manage component-level data.

```
import { useState } from 'react';

function Counter() {
  const [count, setCount] = useState(0);

  return (
    <>
      <p>Count: {count}</p>
      <button onClick={() => setCount(count + 1)}>Increment</button>
    </>
  );
}
```

`useState` is a React Hook for state management.

7. useEffect Hook

Used for side effects like fetching data or DOM updates.

```
import { useEffect } from 'react';

useEffect(() => {
  console.log("Component mounted");
}, []);
```

Empty dependency array means it runs once (like `componentDidMount`).

You can also use it for cleanup.

8. Event Handling

```
function ClickMe() {
  const handleClick = () => alert("Button clicked!");

  return <button onClick={handleClick}>Click Me</button>;
}
```

9. Conditional Rendering

```
{isLoggedIn ? <p>Welcome back!</p> : <p>Please log in</p>}
```

Or using short-circuit:

```
{user && <p>Hello, {user.name}</p>}
```

10. Lists and Keys

```
const users = ['Alice', 'Bob'];
```

```
<ul>
  {users.map((user, index) => (
    <li key={index}>{user}</li>
  ))}
</ul>
```

Always provide a unique key when rendering lists.

11. Forms in React

```
function MyForm() {
  const [name, setName] = useState("");

  const handleSubmit = (e) => {
    e.preventDefault();
    console.log(name);
  };

  return (
    <form onSubmit={handleSubmit}>
      <input value={name} onChange={(e) => setName(e.target.value)} />
      <button type="submit">Submit</button>
    </form>
  );
}
```

12. React Router (Basic)

```
npm install react-router-dom
```

```
import { BrowserRouter, Routes, Route } from 'react-router-dom';
```

```
<BrowserRouter>
  <Routes>
    <Route path="/" element={<Home />} />
    <Route path="/about" element={<About />} />
  </Routes>
</BrowserRouter>
```

13. Styling in React

CSS file: `import './App.css'`

Inline style:

```
<div style={{ color: 'red', fontSize: '20px' }}>Hello</div>
```

CSS modules or styled-components for scoped styles.